ON-BOARD DIAGNOSTIC SYSTEM PID/DATA MONITOR FUNCTION [TCM (CW6A-EL)]

id0502p4280600

Purpose/Function

- The PID/DATA monitor function can monitor control data such as input signals/internal calculation values/output signals using the M-MDS.
- The module control condition can be verified in real-time by the PID/DATA monitor function.

Construction/Operation

PID/DATA monitor item table

—: Not applicable

Item	Unit/Condition	Definition
APP	%	Accelerator pedal position
APP1	%	APP sensor No.1
DGP_DIS_1	km {mile}	Traveled distance after differential protection control operated
DGP_DIS_2	km {mile}	Traveled distance after DGP_MAX_DIF updated
DGP_MAX_DIF	RPM	Maximum rotation difference after differential protection control operated due to excessive rotation difference between left/right drive wheels
DGP_SPD	KPH {MPH}	Vehicle speed with trailing wheels after differential protection control operated due to excessive rotation difference between left/right drive wheels
ECT	°C {°F}	Engine coolant temperature
ECU_A	°C {°F}	ECU internal temperature A
ECU_B	°C {°F}	ECU internal temperature B
ECU_C	°C {°F}	ECU internal temperature C
EOP_DUTY	%	Electric AT oil pump duty cycle
EOP_RLY	Off/On	Electric AT oil pump relay condition
GEAR_RA	Ratio	Gear ratio
GEAR_SEL	1/2/3/4/5/6	Gear shift position
HI_TEMP	—	ATF high temperature mode determination amount
HTM_DIS	km {mile}	Travel distance since determination of ATF high temperature mode
I-Stop_EST	Off/On	i-stop enable state on TCM
LINEDES	kPa {kgf/cm^2, psi}	Target modifier pressure/Target pressure control solenoid pressure
LN_C_CLUTCH	kPa {kgf/cm^2, psi}	Hydraulic control learning value (engaged)
LN_O_CLUTCH	kPa {kgf/cm^2, psi}	Hydraulic control learning value (released)
LN_OV_SCOPE	—	Hydraulic control learning value (mis-learning)
LN_T_CLUTCH	kPa {kgf/cm^2, psi}	Hydraulic control learning value (torque converter)
LOCK_UP	OFF/SLIP/ON	Torque converter (TCC condition)
LONGI_ACCEL	—	Longitudinal acceleration
OP_SW1	Off/On	Oil pressure switch No.1 condition
OP_SW1_OFF	kPa {kgf/cm^2, psi}	After performing on-board diagnostic test mode
OP_SW1_ON	kPa {kgf/cm^2, psi}	After performing on-board diagnostic test mode
OP_SW2	Off/On	Oil pressure switch No.2 condition
OP_SW2_OFF	kPa {kgf/cm^2, psi}	After performing on-board diagnostic test mode
OP_SW2_ON	kPa {kgf/cm^2, psi}	After performing on-board diagnostic test mode
OP_SW3	Off/On	Oil pressure switch No.3 condition
OP_SW3_OFF	kPa {kgf/cm^2, psi}	After performing on-board diagnostic test mode
OP_SW3_ON	kPa {kgf/cm^2, psi}	After performing on-board diagnostic test mode
OP_SW4	Off/On	Oil pressure switch No.4 condition
OP_SW4_OFF	kPa {kgf/cm^2, psi}	After performing on-board diagnostic test mode
OP_SW4_ON	kPa {kgf/cm^2, psi}	After performing on-board diagnostic test mode
OSS	RPM	Output shaft speed
PADDLE_DOWN*1	Off/On	Down switch condition (steering shift switch)
PADDLE_UP*1	Off/On	Up switch condition (steering shift switch)
PUMP_SPEED	RPM	Motor pump unit actual speed
RPM	RPM	Engine speed/min
SC_STATE	Not Active/Active	Shift control execution condition

Item	Unit/Condition	Definition
SE_TYPE	No valid data/Bf_1st Af_2nd/Bf_1st Af_3rd/Bf_1st Af_4th/Bf_1st Af_5th/Bf_1st Af_6th/Bf_2nd Af_1st/Bf_2nd Af_3rd/Bf_2nd Af_4th/Bf_2nd Af_5th/Bf_2nd Af_6th/Bf_3rd Af_1st/Bf_3rd Af_2nd/Bf_3rd Af_4th/Bf_3rd Af_5th/Bf_3rd Af_6th/Bf_4th Af_1st/Bf_4th Af_2nd/Bf_4th Af_3rd/Bf_4th Af_5th/Bf_4th Af_6th/Bf_5th Af_1st/Bf_5th Af_2nd/Bf_5th Af_3rd/Bf_5th Af_4th/Bf_5th Af_6th/Bf_6th Af_1st/Bf_6th Af_2nd/Bf_6th Af_3rd/Bf_6th Af_4th/Bf_6th Af_5th	Gear shift position before shifting gears
SHIFT_CTRL	DEFAULT/ MANUAL/ C_CONTROL/ HIGH_TEMP/ D_MANUAL/ FAIL_SAFE	Shift control mode
SS_ON-OFF	Off/On	On/off solenoid condition
SS1	A	Shift solenoid No.1 actual current
SS1_C	A	Shift solenoid No.1 target current
SS2	A	Shift solenoid No.2 actual current
SS2_C	A	Shift solenoid No.2 target current
SS3	A	Shift solenoid No.3 actual current
SS3_C	A	Shift solenoid No.3 target current
SS4	A	Shift solenoid No.4 actual current
SS4_C	A	Shift solenoid No.4 target current
SSLU	A	TCC control solenoid actual current
SSLU_C	A	TCC control solenoid target current
SSLU_PRES	kPa {kgf/cm^2, psi}	TCC control solenoid target pressure
SSP	A	Pressure control solenoid actual current
SSP_C	A	Pressure control solenoid target current
SSP_PRES	kPa {kgf/cm^2, psi}	Pressure control solenoid target pressure
TFT	°C {°F}	ATF temperature
THOP	%	Throttle opening angle
TORQUE_ACT	Nm	Actual engine torque
TORQUE_DES	Nm	Desired engine torque
TR	P/SHIFT_DOWN/ SHIFT_UP/M/D/N/ R	Range position
TS_OD	km {mile}	Traveled distance when AT slip occurred

Item	Unit/Condition	Definition
TS_SE	No valid data/Bf_1st Af_2nd/Bf_1st Af_3rd/Bf_1st Af_4th/Bf_1st Af_5th/Bf_1st Af_6th/Bf_2nd Af_1st/Bf_2nd Af_3rd/Bf_2nd Af_4th/Bf_2nd Af_5th/Bf_2nd Af_6th/Bf_3rd Af_1st/Bf_3rd Af_2nd/Bf_3rd Af_4th/Bf_3rd Af_5th/Bf_3rd Af_6th/Bf_4th Af_1st/Bf_4th Af_2nd/Bf_4th Af_3rd/Bf_4th Af_5th/Bf_4th Af_6th/Bf_5th Af_1st/Bf_5th Af_2nd/Bf_5th Af_3rd/Bf_5th Af_4th/Bf_5th Af_6th/Bf_6th Af_1st/Bf_6th Af_2nd/Bf_6th Af_3rd/Bf_6th Af_4th/Bf_6th Af_5th	Gear position before/after shifting when AT slip occurred
TS_VS	KPH {MPH}	Vehicle speed before/after shifting when AT slip occurred
TSS	RPM	Turbine/input shaft speed
UPSHIFT_REV	Off/On	Shift-up record due to engine request
VPWR	V	TCM power supply voltage
VSS	KPH {MPH}	Vehicle speed

*1 : With steering shift switch